CRITERIA FORCING THE PUSHFORWARD OF ABELIAN SHEAVES ALONG A MORPHISM OF TOPOI TO BE EXACT

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites two categories sections, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Right adjoints are left exact and exactness criteria for functors. Stacks Project, Categories, Section categories-section-exact-functor (tag 0034) and Section categories-section-adjoint (tag 0036). Supplies the left exactness of f_* and the framework for reducing exactness to preservation of surjections.

Source reference. Original reference 0033: sources/stacks-chow/tags.txt, source lines 120–120. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0036: sources/stacks-chow/tags.txt, source lines 123–123. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 010M: sources/stacks-chow/tags.txt, source lines 1332–1332. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04D6: sources/stacks-chow/tags.txt, source lines 5716–5716. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : Sh(\mathcal{C}) \to Sh(\mathcal{D})$ be a morphism of topoi. Assume at least one of the following properties holds*

(1) *f_* transforms surjections of sheaves of sets into surjections,*
(2) *f_* transforms surjections of abelian sheaves into surjections,*
(3) *f_* commutes with coequalizers on sheaves of sets,*
(4) *f_* commutes with pushouts on sheaves of sets,*

Then $f_* : Ab(\mathcal{C}) \to Ab(\mathcal{D})$ is exact.

Proof. Since $f_* : Ab(\mathcal{C}) \to Ab(\mathcal{D})$ is a right adjoint we already know that it transforms a short exact sequence $0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$ of abelian sheaves on $\mathcal{C}$ into an exact sequence

$$0 \to f_*\mathcal{F}_1 \to f_*\mathcal{F}_2 \to f_*\mathcal{F}_3$$

see Categories, Sections 0033 and 0036 and Homology, Section 010M. Hence it suffices to prove that the map $f_*\mathcal{F}_2 \to f_*\mathcal{F}_3$ is surjective. If (1), (2) holds, then this is clear from the definitions. By Sites, Lemma 04D6 we see that either (3) or (4) formally implies (1), hence in these cases we are done also. □

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